But as raster graphics (as the PFD in figure 2.5 and the P&ID in figure 2.6 are printed here), a computer cannot understand the information contained in these documents without some further processing. Luckily, computer vision, and more recently, vision capable *large language models (LLMs)* can help to digitize these documents into a structured format that can be used.

TRY IT OUT: EXTRACT INFORMATION FROM AN ENGINEERING DOCUMENT

P&IDs, as shown in Figure 2.6, often exist as legacy documents, either in paper form or scanned PDF documents with multiple revisions, annotations, and different symbols. The diagrams themselves are designed for human, rather than machine, interpretation and contain important information related to industrial sensor locations and identifiers, control logic, and topological information. Digitizing these documents and transforming the information they contain into structured information that can be processed by machines is critical in any digital twin of an industrial process.

LLMs with multi-modal understanding can be used to analyze engineering documents and transform them into machine readable representations that can further be used to create a digital model of the process defined in the diagram.

Listing 2.4 uses Anthropic's Claude Sonnet 4 model and a simple prompt to convert the sample P&ID diagram shown in figure 2.6, to a structured JSON format. Fine-tuning the prompt allows you to customize the structure of the generated JSON. To run this code, you will need an API key from Anthropic, which you can get here: https://www.anthropic.com/api. The sample P&ID is provided as an image in the books GitHub repository.